

Colour Mixing Light Show

Session 1: Discover the Magic of RGB!

| The Secret of White Light

What if I told you that the colour **WHITE** is actually made of all the colours mixed together?

You have seen beautiful lights during Diwali or colourful spotlights on your school stage. Today, we find out the secret behind those colours!

The Magic Trio

In the world of light, everything starts with three "Magic" primary colours. You cannot make these by mixing anything else!

- **RED**
- **GREEN**
- **BLUE**

Call them out together: **RED! GREEN! BLUE!**

| Where is the Magic?

We see these magic colours all around us in India!

- **Diwali Decorations:** Tiny LED bulbs on homes in Delhi or Mumbai glow in red, green, or blue light independently to make the city sparkle!
- **Traffic Signals:** From Bangalore to Kolkata, every traffic light uses big, bright Red and Green lights to tell cars when to stop and go!
- **Your Screen:** Look at a phone or TV screen. It actually uses millions of tiny Red, Green, and Blue dots to show you all your favorite cartoons!

The Colour Mixing Show

What happens when two primary colours of light meet? It's like a magic trick!

- Red + Green = **YELLOW!** (Like stage spotlights during school functions)
- Red + Blue = **MAGENTA!** (A bright pinkish-purple)
- Green + Blue = **CYAN!** (A bright blue-green)

Guess First! What do you think happens when we mix ALL THREE?

The Big Reveal: White Light

When Red, Green, and Blue light all come together, they create pure **White Light!**

- **Indian Example:** The white LED bulbs and tube lights in your home or classroom are actually mixing these colours right now!
- **What about Darkness?** When there is NO light, we see Black. Think of a power cut at night—it goes dark because there is zero light!

Light is Different from Paint!

Don't be confused! Mixing light is a bit like magic, while mixing paint is like cooking.

Mixing Paint (Like Camlin sets!)

When you mix paints, the colour gets **DARKER.**

Example: Blue + Yellow paint = Green.

Mixing Light (The Magic Way!)

When you mix light, the colour gets **BRIGHTER!**

Example: Red + Green light = Yellow.

Remember: Light adds more brightness!

You are a Colour Scientist!

Let's Recap!

- Primary Colours of Light: Red, Green, Blue.

- Red + Green = Yellow.
- All three together = White Light.

Success Criteria

- I can name the 3 magic colours of light.
- I can predict what happens when lights mix.
- I can name one place in India where coloured lights glow.
- I participated in the light mixing activity!

CONGRATULATIONS!

You have finished Session 1 of the Light Show!